

Module 3 — Health Expenses & Insurance Claims

(modules/health)

Purpose

Centralize medical expenses and their reimbursement lifecycle across Portuguese private insurers (Fidelidade, Allianz, Médis), mutual funds ("caixas"), and public subsidies (ADSE-style), computing out-of-pocket costs and automating claim matching to bank credits. Supports per-family-member coverage plans with coordination-of-benefits rules and deadline tracking per insurer.

Actors & User Stories

- *As a household manager*, I log a medical receipt, and the system suggests which insurer(s) to claim from based on coverage.
- *As a claimant*, I upload a receipt once and receive automatic reminders if payment is pending past the insurer's SLA.
- *As a dependent*, my medical expenses are tracked under my name but rolled up in household views.
- *As a tax preparer*, I export all health expenses marked as IRS-deductible ("despesas de saúde") for e-fatura submission.
- *As an auditor*, I trace the complete chain: receipt → expense → claim(s) → bank credit(s) and see why each part was accepted/rejected.

Data Model

All monetary fields (`_eur` suffix) are **NUMERIC(10,2)** decimal EUR amounts (e.g. **19.99**), never minor-unit integers; UI displays them pt-PT-formatted (e.g. **19,99€**).

- **MedicalExpense**: `id`, `entity_id`, `member_id`, `incurred_date`, `provider_nif` (tax ID), `provider_merchant_id`, `category`(APPOINTMENT|DIAGNOSTIC|PHARMACY|DENTAL|OPTICAL|MENTAL_HEALTH|HOSPITAL|OTHER), `description`, `gross_amount_eur`, `copay_amount_eur?`, `document_id`, `transaction_id?`, `is_irs_deductible`, `tags[]`.
- **Claim**: `id`, `medical_expense_id`, `insurer_id`, `status`(INCURRED|SUBMITTED|PARTIALLY_REIMBURSED|FULLY_REIMBURSED|CLOSED|REJECTED), `submitted_at`, `submission_reference?`, `expected_amount_eur`, `submitted_amount_eur`, `deadline_for_payment`, `last_status_update_at`, `rejection_reason?`, `tags[]`.
- **ClaimReimbursement** (one per payout, enabling partial/staged reimbursements): `id`, `claim_id`, `reimbursement_date`, `received_amount_eur`, `received_by_transaction_id`, `source_document_id`, `is_final`, `notes`.
- **Insurer**: `id`, `name`, `kind`(INSURANCE|MUTUAL_FUND|PUBLIC_SUBSIDY), `nif`, `contact_email`, `website`, `sla_days`, `is_deleted`, `logo_document_id`.
- **CoveragePlan**: `id`, `member_id`, `insurer_id`, `plan_name`, `coverage_start`, `coverage_end`, `eligible_categories[]`, `deductible_eur`, `copay_pct?`, `max_out_of_pocket_annual_eur?`, `coordination_of_benefits_order`, `notes`.

- **PrescriptionItem** (recurring/standing treatments): `id`, `member_id`, `provider_nif`, `name`, `dosage`, `frequency(DAILY|WEEKLY|MONTHLY)`, `start_date`, `end_date`, `copay_eur?`, `estimated_annual_cost_eur`, `linked_claims[]`.
- Derived: `out_of_pocket_eur` = `gross` - $\sum$ `reimbursed` (respects deductible and CoB cap); `reimbursed_pct` = $\sum$ `reimbursed` / `gross`. All Claim transitions logged in `AuditLog`; `ClaimReimbursement` links to the delivering bank `Transaction`.

Functional Requirements

- **FR-3.1 Claim Lifecycle State Machine:** `INCURRED` → `SUBMITTED` → `PARTIALLY_REIMBURSED` → `FULLY_REIMBURSED/CLOSED` (+ `REJECTED`). Rejected claims revert the expense to fully out-of-pocket and may be resubmitted. Illegal transitions (e.g., `CLOSED` → `SUBMITTED`) rejected by the API.
- **FR-3.2 Multi-Insurer Stacking & Coordination of Benefits:** multiple claims per expense across insurers; `coordination_of_benefits_order` defines claim sequence (primary insurer → secondary mutual fund → tertiary public subsidy). $\sum$ all claims cannot exceed `gross_amount_eur` minus deductible. Each subsequent insurer's share is computed net of prior reimbursements.
- **FR-3.3 Coverage Eligibility & Copay:** match expense category against `CoveragePlan.eligible_categories`; apply copay (flat or %) and deductible per plan; track annual out-of-pocket max; suggest applicable plans; flag out-of-coverage-window expenses.
- **FR-3.4 Deadline & SLA Tracking:** `Claim.deadline_for_payment` = `submitted_at` + `insurer.sla_days`; Alert Engine triggers overdue reminders; historical SLA compliance tracked per insurer.
- **FR-3.5 NIF & Provider Tax ID Capture:** store `provider_nif` on `MedicalExpense` and `Insurer.nif` for e-fatura matching and audit.
- **FR-3.6 Prescription & Recurring Treatment Plans:** `PrescriptionItem` tracks standing treatments; auto-create `MedicalExpense` records on cadence (Celery Beat); link multiple claims to the same prescription.
- **FR-3.7 Bank Credit Reconciliation:** propose `ClaimReimbursement` links by matching bank credits to open/partial claims (date within $\pm$ SLA_days of deadline, amount $\pm$ 5%, insurer reference). High-confidence auto-confirmed; others → review.
- **FR-3.8 IRS Export for Despesas de Saúde:** flag `is_irs_deductible`; export CSV/XML with household NIF, provider NIF, amount, date for e-fatura integration.

Automation Rules

- **Insurer Suggestion:** on new expense, scan member `CoveragePlan` records; suggest eligible insurers by category and active window.
- **CoB Ordering:** on confirming a claim, advance to next insurer in `coordination_of_benefits_order`; auto-adjust expected reimbursement based on prior payouts.
- **Overdue Alerts:** daily Celery Beat checks claims with `deadline_for_payment` < `today` and `status` $\neq$ `CLOSED|FULLY_REIMBURSED`; create alert.
- **Partial Accumulation:** each `ClaimReimbursement` updates `Claim.status` (engine decides `FULLY_REIMBURSED` vs stays `PARTIALLY_REIMBURSED` per `is_final`).

UI / Screens

- **Expense Entry:** date, provider (search by name/NIF), category, gross/copay amounts, coverage plan selector, attachment, IRS-deductible checkbox.
- **Claims Kanban:** swimlanes by status; card shows expense summary, insurer, submitted date, deadline, expected vs received.
- **Reimbursement Matcher:** side-by-side open claims vs unmatched bank transactions; suggest + confirm.
- **Member & Insurer Summaries:** per-member out-of-pocket YTD, per-insurer reimbursement rate + SLA compliance, prescription status.
- **Overdue Dashboard:** claims past deadline with action buttons.

API Surface

- **POST** /medical-expenses (with auto-insurer suggestion) · **GET** /medical-expenses (filters) · **PATCH** /medical-expenses/{id}.
- **POST** /claims · **GET** /claims (status, member, insurer, overdue) · **POST** /claims/{id}/transition (legal-transition validation).
- **POST** /claims/{id}/reimbursements · **GET** /claims/{id}/reimbursements.
- **CRUD** /coverage-plans · **CRUD** /insurers · **CRUD** /prescriptions.
- **GET** /health/irs-export · **GET** /health/analytics.

Analytics & KPIs

- Out-of-pocket over time (by member, category, insurer status).
- Reimbursement rate per insurer; SLA compliance (% paid on time).
- Per-member annual out-of-pocket vs deductible and max thresholds.
- Spend by category & coverage gap (low-reimbursement categories → plan upgrade opportunities).
- Claims leakage: count of INCURRED expenses not yet submitted + value at risk.

Edge Cases & Validation

- **Over-Reimbursement Guard:** $\sum \text{ClaimReimbursement.received}$ across all claims for one expense $\leq \text{gross_amount_eur}$; second claim auto-adjusted; flag for review with override + documented reason.
- **Partial Accumulation:** `is_final=false` does not close claim; `is_final=true` triggers state check.
- **Rejected Reopen:** rejecting resets `Claim.status` to `INCURRED`; expense reverts to full out-of-pocket until resubmitted or closed.
- **Duplicate Receipt Prevention:** SHA-256 + provider NIF + amount + date; duplicates flagged for merge.
- **Prescription Cadence Validation:** frequency + dates must be logical; missed/skipped months flagged.
- **Provider NIF Format:** Portuguese 9-digit check-digit validation.

Additional / Enriched Requirements

- **Preset Insurers (seeded):** Fidelidade, Allianz, Médis, Multicare, Seguros Tranquilidade, ADSE, SNS-based reimbursements.
- **Mutual Funds ("Caixas"):** employer/union schemes tracked separately from commercial insurers.
- **Per-Plan Deductible** applied once per calendar year; **annual out-of-pocket cap** → 100% coverage for remainder of year once met.

- **Stacking example:** €100 expense → Insurer A covers €80 (OOP €20); Insurer B sees €20, covers €15 net of its €5 deductible → household OOP €5.
- **Recurring Treatments Auto-Submission:** monthly prescriptions may auto-create **Claim** in **SUBMITTED** with expected reimbursement from prior cadence (user reviews/overrides).
- **IRS Compliance:** "Despesas de Saúde" partially deductible above a threshold; export includes NIFs, date, amount, category.
- **Full Chain Traceability:** bank transaction → reimbursement → claim → expense → original receipt document.

Open Questions / Decisions

1. **Auto-submit new claims?** → *Require confirmation; auto-submission opt-in per insurer.*
2. **Prescription auto-creation cadence job time?** → *Midnight (avoids lag).*
3. **Deductible reset date?** → *Per coverage plan anniversary; default Jan 1.*
4. **Over-reimbursement handling?** → *Flag to user; allow override with documented reason (fraud check).*
5. **Provider NIF capture?** → *Mandatory for professional providers; optional for walk-in clinics.*

Definition of Done

- State machine enforced; illegal transitions rejected; all transitions logged.
- Multi-insurer math tested: CoB ordering, deductible, out-of-pocket-max, no over-reimbursement.
- Bank reconciliation: ClaimReimbursement linked to Transaction; SLA compliance tracked.
- Overdue alerts: Celery job runs; Review Queue receives alerts.
- Prescription cadence: auto-creates expenses + claims; no failures on month-end edge dates.
- IRS export: includes flagged expenses, validates NIF formats, correct sum.
- Analytics queries sub-100ms for typical household (100–500 expenses, 5–10 insurers).
- Frontend: Kanban, matcher, summaries, alerts; all tests pass.
- OpenAPI complete; data model in ERD.

Integration Contract

- **Exposes:** **MedicalExpense**, **Claim**, **ClaimReimbursement** as link targets for Banking (M2) and Assets (M6); out-of-pocket totals feed Dashboards.
- **Consumes:** **Merchant** (provider validation), **Entity**/members, **Category** (health sub-tree), **Document** (receipts), **Transaction** (M2), **Tag**, **User** (attribution/audit).
- **Depends on:** Notification Engine (overdue alerts), Reconciliation Engine (bank-match proposals), Intelligence Provider (OCR when needed).